LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

Anonymous submission

Abstract

Multi-center medical models are often trained asynchronously across hospitals with different devices, patient populations, and diagnostic label coverage. This paper studies one-shot asynchronous medical model merging, where each client uploads once after local training and the server constructs a global model without raw medical images or federated optimization rounds. We identify aggregation-induced predictive collapse as a central failure mode of generic post-hoc merging under incomplete local diagnostic coverage. We also show that raw accuracy can hide this failure because long-tailed medical priors make majority-class collapse deceptively competitive. We propose LAMP-Merge, a long-tail-aware medical prototype merging framework that reconstructs a diagnostic prototype head from client-side class statistics and calibrates long-tailed prevalence with a bounded prior bias. Across five medical datasets, five model families, three client counts, and three non-IID skew levels, LAMP-Merge wins or ties the strongest non-LAMP baseline in 68 out of 75 client-average cells, while also improving macro recall, macro F1, and prediction-collapse diagnostics.

I INTRODUCTION

Modern medical artificial intelligence is increasingly developed in a multi-center environment. A useful diagnostic model may require information from several hospitals, scanners, departments, and patient populations, but these participants are rarely synchronized. In practice, hospitals finish local training at different times, use devices with different imaging characteristics, and observe different subsets of the diagnostic label space. For example, a hematology center may contain abundant common blood-cell categories but very limited rare cell types, while an ultrasound center may be dominated by a small number of routine findings. These constraints motivate one-shot asynchronous medical model merging: after local training, each hospital uploads once, and a server constructs a global diagnostic model without iterative communication.

This setting is different from federated learning. Federated optimization assumes repeated rounds of server aggregation, client-side continued training, and re-uploaded updates. Such a protocol is expensive and often unrealistic when clients are asynchronous, offline, or unwilling to participate in repeated training rounds. Our setting is also different from standard post-hoc model merging in natural language processing, where models are usually fine-tuned from the same foundation model on semantically rich tasks with broad label coverage. In medical imaging, client data are often label-incomplete and long-tailed: each client can be locally meaningful, but no single client necessarily covers the entire diagnostic space.

A direct application of existing post-hoc merging algorithms exposes a failure mode that is easy to miss if only accuracy is considered. We observe aggregation-induced predictive collapse: local medical models may still predict multiple categories before merging, but many parameter-space merging rules turn the merged model into a single-class or few-class predictor. On our diagnostic probe, generic merging baselines excluding LAMP-Merge obtain an average collapse ratio of 0.7993 and use only 1.8600 effective predicted classes. Standard averaging alone has a collapse ratio of 0.8028. In contrast, a natural-image CIFAR-10 partial-label control with standard averaging remains non-collapsed, with a collapse ratio of 0.2425 and 10.0000 effective predicted classes. This suggests that medical long-tailed class priors and incomplete local diagnostic coverage amplify a distinct post-hoc merging failure mode.

We propose LAMP-Merge, a Long-tail-Aware Medical Prototype merging framework for asynchronous medical model merging. LAMP-Merge keeps the communication one-shot and does not require the server to access raw medical images. Each client computes class-level diagnostic statistics locally and uploads only the checkpoint interface information, class support counts, class prevalence counts, and reference-backbone class prototypes. The first module reconstructs a global diagnostic prototype head in a shared reference feature space, preserving per-class local expertise rather than averaging incompatible classifier heads. The second module applies bounded long-tail prevalence calibration when the uploaded class prevalence indicates a dominant diagnosis class. This design follows two empirical observations: post-hoc merging induces predictive collapse in medical images, and, under strongly long-tailed medical priors, collapse can appear beneficial in accuracy because the majority class itself occupies a large portion of the test distribution.

Our contributions are summarized as follows. First, we formulate a one-shot asynchronous medical model merging problem that is distinct from federated learning and from data-free NLP-style model merging. Second, we identify predictive collapse as a central failure mode of generic merging algorithms under incomplete local diagnostic coverage. Third, we introduce LAMP-Merge, which uses client-side class prototypes and prevalence counts to reconstruct a non-collapsed global diagnostic model without exposing raw images. Fourth, we provide a broad empirical study over five medical image datasets, five model families, three client counts, and three non-IID skew levels; LAMP-Merge wins or ties the strongest non-LAMP baseline in 68 out of 75 client-average cells.

II RELATED WORKS

Generic Post-hoc Model Merging

Post-hoc model merging combines trained checkpoints without jointly retraining all participating models. Existing methods can be grouped by the variables on which merging is performed. The first group performs direct interpolation in weight space. Weight averaging and model soups average checkpoints and are effective when the models lie in a compatible basin (Wortsman et al. 2022). The second group performs arithmetic in update space. Task arithmetic represents a fine-tuned model by its parameter displacement from a shared initialization and merges such displacements linearly (Ilharco et al. 2023). The third group attempts to reduce parameter interference before averaging. Git Re-Basin aligns permutation-equivalent neurons (Ainsworth, Hayase, and Srinivasa 2023), while TIES-Merging and DARE-style methods trim small updates, resolve sign conflicts, or randomly drop and rescale task vectors (Yadav et al. 2023; Yu et al. 2024). The fourth group introduces auxiliary geometric or statistical structure, including curvature-aware Fisher merging, RegMean, Model Stock, Breadcrumbs, Iso-C, FreeMerge, and RobustMerge (Matena and Raffel 2022; Jin et al. 2023).

Although these methods provide strong generic baselines, their fusion variables are not diagnostic-class aware. They operate on global parameters, task vectors, curvature estimates, or update signs, but they do not condition the merge on whether client i contains reliable evidence for diagnosis class c. This assumption is restrictive in multi-center medical imaging, where local training sets are often label-incomplete and long-tailed. When $n_{i,c} = 0$ for many client-class pairs, class-agnostic fusion can average a reliable class direction with absent or conflicting directions from other clients. The resulting checkpoint may preserve the dominant-class direction while suppressing rare or locally missing diagnostic classes. Our prediction-distribution analysis shows this behavior as aggregation-induced predictive collapse: generic baselines can achieve nontrivial raw accuracy while producing low balanced accuracy, low macro F1, and a highly concentrated predicted-class distribution. LAMP-Merge addresses this failure by changing the unit of fusion from a global parameter update to class-conditioned medical evidence, represented by client-side diagnostic prototypes and class prevalence counts.

Federated and Decentralized Medical Learning

Federated and decentralized learning study collaborative training under privacy constraints. Federated averaging and its variants repeatedly broadcast a global model, perform local client optimization, and aggregate uploaded updates. Swarm learning removes the central parameter server and coordinates training through decentralized mechanisms (Warnat-Herresthal et al. 2021). Gossip-style learning further propagates models asynchronously across participants. These protocols are appropriate when participating hospitals can remain online for multiple communication rounds and can continue optimizing in response to intermediate global models.

The protocol studied in this paper is different. We assume that hospitals finish local training independently and upload once. The server does not send a sequence of global checkpoints to clients, and clients do not perform additional optimization after observing server-side aggregates. Therefore, federated and decentralized learning solve a different problem: multi-round collaborative training. LAMP-Merge instead targets one-shot asynchronous diagnostic checkpoint fusion. It preserves the privacy and deployment advantages of local training while avoiding the operational cost of repeated server-client interaction.

Medical Fusion and Multimodal Clinical Modeling

Medical fusion methods exploit structures that are specific to clinical data. Representative examples include continuous-time attention for irregularly sampled clinical observations, graph-based fusion of physiological sensors, and image-EHR fusion for chest X-ray plus clinical time-series prediction (Shukla and Marlin 2021; Zhang et al. 2022; Hayat, Geras, and Shamout 2022). These studies demonstrate that medical data are not merely another application domain of generic fusion; their temporal irregularity, modality heterogeneity, clinical measurement process, and disease prevalence structure must be modeled explicitly.

However, these methods do not solve the checkpoint-merging problem considered here. They are usually trained with access to patient-level samples, synchronized or partially synchronized multimodal records, and task feedback during model learning. Such assumptions are incompatible with a server that cannot inspect raw medical images, sample-level features, sample-level logits, or validation predictions. Moreover, these methods fuse input modalities or patient representations, rather than independently trained image classifiers. LAMP-Merge inherits the medical principle that fusion should respect domain structure, but instantiates it in a one-shot checkpoint-merging protocol by using only class-level diagnostic statistics computed on the client side.

Medical Model Merging

Recent medical model merging studies are closer to our setting, but they still rely on different communication or data assumptions. FedSoup is a personalized federated learning framework: during federated training, each client maintains a model pool and selects interpolations between local and global models for personalization (Chen et al. 2023). It therefore depends on federated training dynamics and does not output a single global checkpoint through one post-hoc upload. MedMerge learns kernel-level fusion weights for transfer learning from models with different initializations (Almakky et al. 2024). Its objective is target-domain adaptation with fusion-weight learning, whereas our objective is to merge same-task client checkpoints when each client may cover only part of the diagnostic label space. FissionFusion generates multiple models and performs hierarchical souping selected by validation performance (Sanjeev et al. 2024). This is effective for medical model generation and selection, but its selection stage depends on validation feedback and model trajectories that are not available under our server-side privacy boundary. MedSAMix merges SAM-based generalist and specialist segmenters through layer-wise optimization for medical segmentation (Yang et al. 2026); it addresses foundation segmentation models rather than class-skewed diagnostic classifiers. InMerge merges kernels within a single model during fine-tuning (Wang et al. 2025), making it an in-model robustness regularizer rather than an inter-client merging protocol.

These methods reveal two limitations for our scenario. First, several approaches require server-side target images, validation feedback, or optimization trajectories to estimate fusion weights or select soups. Centralizing such empirical feedback conflicts with the requirement that the server must not access raw medical data or sample-level outputs. Sending the evaluation process back to clients would introduce an additional feedback protocol and move the method toward federated optimization rather than one-shot merging. Second, existing medical merging methods do not explicitly model class-dependent diagnostic availability. They may account for medical domain shift, transfer learning, segmentation specialization, or robustness, but they do not represent the fact that a client can be highly reliable for one diagnosis class and uninformative for another. LAMP-Merge is designed for this missing regime: it performs a single server-side merge from uploaded checkpoints, class support counts, class prevalence counts, and reference-space class prototypes, thereby aligning the fusion variables with the diagnostic structure of multi-center medical imaging.

III LAMP FRAMEWORK

Problem Formulation

Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Each client trains a local model independently and asynchronously. After local training, the client performs one upload containing the model interface and class-level diagnostic statistics computed locally. The server must produce one global model in a single post-hoc merging step. The server cannot access raw medical images, sample-level features, sample-level logits, or sample-level predictions.

The key difficulty is incomplete local diagnostic coverage. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c. In medical partitions, $D_{i,c}$ can be empty for many classes. Thus, a client can be reliable for a subset of diagnoses but unreliable for unseen classes. A class-agnostic parameter merge has no explicit mechanism to preserve this local reliability structure.

Observation 1: Aggregation-induced Predictive Collapse

Before introducing the algorithm, we formalize the diagnostic quantity used to identify collapse. For a model evaluated on N test images, let

$$q(c) = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}[\hat{y}_n = c]$$

be the predicted class distribution. We define the collapse ratio as

$$\rho = \max_c q(c).$$

A model with ρ close to one behaves like a single-class predictor.

Our first observation is aggregation-induced predictive collapse: in local-class-coverage medical settings, post-hoc parameter merging can transform multiple non-degenerate local predictors into a globally collapsed predictor. The local clients may be biased because they see only part of the diagnostic space, but they can still respond to multiple classes. The collapse mainly appears after class-agnostic aggregation, where parameter averaging, sign voting, or pruning amplifies locally dominant directions without preserving per-class evidence.

Observation 2: Collapse Benefit under Long-tailed Priors

Our second observation is that collapse can be deceptively useful under long-tailed medical priors. Let

$$p_{\max} = \max_c P(y = c)$$

be the highest test-set class prior. A model that always predicts the most frequent class obtains

$$\mathrm{Acc}_{\text{collapse}} = p_{\max}.$$

Therefore, when the most common diagnosis occupies a large fraction of the distribution, a collapsed predictor may achieve high accuracy without real multi-class diagnostic ability. This explains why some baselines can look competitive in raw accuracy while failing balanced accuracy, macro F1, and prediction-distribution diagnostics. A useful medical merging method should avoid losing rare-class discrimination, while still respecting genuine prevalence information when the task is strongly long-tailed.

M1: Diagnostic Prototype Reconstruction

LAMP-Merge first reconstructs a class-wise diagnostic head in a shared reference feature space. Let ϕ_0 be the frozen reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c, client i computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y=c]\phi_0(T(x)),$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

The server converts support counts into evidence weights:

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0],$$

where $\gamma = 0.45$ is the default evidence exponent. The normalized reliability of client i for class c is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^{K} e_{j,c}}.$$

The global diagnosis prototype is then

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}.$$

Finally, LAMP-Merge synthesizes a cosine prototype classifier:

$$w_c = s\frac{p_c}{\|p_c\|_2},$$

where $s = 20$ is the default head scale. This module directly addresses Observation 1: instead of averaging incompatible classifier heads, it preserves the classes for which each client has evidence and reconstructs a global multi-class diagnostic head.

M2: Long-tail Prevalence Calibration

M1 prevents class-agnostic aggregation from collapsing the classifier, but medical tasks may contain real long-tailed prevalence. To account for Observation 2, each client also uploads class prevalence counts $m_{i,c}$. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^{K} m_{i,c}}{\sum_{k=1}^{C}\sum_{i=1}^{K} m_{i,k}}.$$

Table 1: Dataset statistics used in the experiments.

Dataset	Classes	Channels	Train	Val	Test
Blood	8	3	11959	1712	3421
Derma	7	3	7007	1003	2005
Organ-C	11	1	12975	2392	8216
Organ-S	11	1	13932	2452	8827
Ultrasound	8	3	3869	549	1113
Total	–	–	49742	8108	23582

We measure dominant-class imbalance by

$$r = C \max_c \pi_c.$$

M2 is activated only when $r > \tau$, where $\tau = 2.5$ by default. When activated, it adds a centered log-prior bias

$$b_c = \lambda\left(\log\pi_c - \frac{1}{C}\sum_{k=1}^{C}\log\pi_k\right),$$

with default maximum calibration strength $\lambda = 6.0$. The final score is

$$\mathrm{score}_c(x) = w_c^{\top}\phi_0(T(x)) + b_c.$$

The bias is centered, bounded, and attached to the prototype classifier; it calibrates prevalence without replacing the multi-class diagnostic directions reconstructed by M1.

IV EXPERIMENTS

Experimental Setup

We evaluate LAMP-Merge under one-shot asynchronous medical model merging. Each client trains locally on its private class-skewed partition. After local training, it uploads one checkpoint and the class-level statistics required by LAMP-Merge. The server merges once and does not run iterative server-client optimization.

Datasets We use five 224-resolution medical image classification datasets. All tasks are single-label multi-class diagnosis or organ recognition tasks. The exact statistics are shown in Table 1.

For standard vision backbones, we evaluate ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny. For vision-language models, we evaluate openai/clip-vit-base-patch32. For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$. All results use seed 42. The benchmark contains 225 raw test-accuracy cells and 75 client-average cells. The client-average metric averages the three β values for each fixed client number, reducing sensitivity to a single partition skew.

Baselines We compare with twelve post-hoc merging baselines: avg, ties, dare_linear, dare_ties, regmean, fisher, breadcrumbs, model_stock, from, iso_c, free_merge, and robustmerge. The main comparison counts a cell as successful when LAMP-Merge is greater than or equal to the strongest non-LAMP baseline in that cell.

Experiment Environment

All experiments were run on a server with two Intel(R) Xeon(R) Silver 4210 CPUs @ 2.20GHz, 40 logical CPU threads in total, and four NVIDIA GeForce RTX 2080 Ti GPUs with 11264 MiB memory each. The NVIDIA driver version is 535.154.05. The operating system is Ubuntu 20.04.1 LTS. The software environment is the conda environment MM, using Python 3.10.20, PyTorch 2.6.0+cu118, torchvision 0.21.0+cu118, NumPy 2.2.6, and CUDA runtime 11.8.

Hyperparameter Settings

Unless otherwise stated, LAMP-Merge uses evidence exponent $\gamma = 0.45$, prototype head scale $s = 20$, long-tail activation threshold $\tau = 2.5$, and prevalence calibration strength $\lambda = 6.0$. These hyperparameters are fixed for all datasets, backbones, client counts, and skew levels.

Main Client-average Results

Tables 2–6 report all client-average results from the full benchmark. Across the 75 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in 68 cells, corresponding to 90.7%. Across all 300 cells including raw and client-average entries, LAMP-Merge wins or ties in 241 cells.

Beyond Accuracy: Recall and Collapse Diagnostics

Because medical diagnosis datasets are often long-tailed, raw accuracy alone is not sufficient to establish a clinically meaningful improvement. A collapsed predictor can obtain high accuracy when the dominant diagnosis class is frequent. Therefore, we explicitly evaluate whether the gain remains under class-balanced metrics. For single-label multi-class classification, balanced accuracy is macro recall:

$$\mathrm{BA} = \frac{1}{C}\sum_{c=1}^{C}\frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}.$$

Thus, a method that only predicts the majority class may have acceptable accuracy but poor macro recall. We further report macro F1, collapse ratio, effective predicted classes, and prediction-true total variation distance to diagnose whether a method preserves multi-class discrimination.

This diagnostic experiment is evaluated on five medical datasets: BloodMNIST, Ultrasound, DermaMNIST, Organ-C, and Organ-S. For each dataset, we use four backbones, ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny, under $K = 3$, $\beta = 0.01$, and seed 42, resulting in 20 dataset-backbone diagnostic cases. On these 20 cases, LAMP-Merge improves accuracy from 0.3156 for the strongest non-LAMP/client reference to 0.5885. More importantly, the improvement is larger under class-balanced metrics: balanced accuracy, i.e., macro recall, increases from 0.1810 to 0.5747, and macro F1 increases from 0.1123 to 0.5352. Its collapse ratio is 0.2294, much lower than 0.7477 for the strongest reference under this metric, and its effective predicted classes increase from 2.1360 to 8.0390. These results show that LAMP-Merge does not merely exploit majority-class accuracy; it also improves recall-oriented and distribution-oriented diagnostics.

The ablation is computed at the same granularity as the main client-average table: for each fixed dataset, backbone, and client count, we first average over the three Dirichlet skew levels and then compare the four controlled variants. M1 is the main source of the gain, while M2 improves long-tail stress cases without serving as an independent merging method. On dermamnist_224/ResNet/$K = 3/\beta = 0.1$, M1 only obtains 0.4314 accuracy, whereas full LAMP-Merge obtains 0.6723.

Table 9 further decomposes the ablation by dataset. Blood, Ultrasound, Organ-C, and Organ-S mainly benefit from M1, indicating that class-wise diagnostic prototype reconstruction is the dominant mechanism across heterogeneous medical image modalities. Derma shows the largest gap between LAMP-Merge and M1 only, which supports the role of M2 as a bounded prevalence calibration under a strong long-tailed diagnostic prior.

Privacy Preservation

LAMP-Merge is not federated learning and does not require multi-round communication. The server receives one upload after local training. The uploaded information consists of the checkpoint, class support counts, reference-backbone class prototypes, and class prevalence counts. The server never receives raw medical images, sample-level features, sample-level logits, or sample-level predictions. Since prototypes and prevalence are aggregated per diagnosis class on the client side, the server operates on class-level summaries rather than patient-level records.

Additional Ablation and Sensitivity Results

We report additional ablation and sensitivity analyses to isolate the functional role of each component. The inter-module ablation is evaluated on the full benchmark, containing five medical datasets, five model families, three client counts, and three non-IID skew levels, i.e., 225 raw accuracy cells and 75 client-average cells. M1 only retains diagnostic prototype reconstruction while removing long-tail prevalence calibration. avg+M2 removes prototype reconstruction and applies

Table 2: Client-average test accuracy on resnet. Each entry averages the three Dirichlet settings for a fixed client number.

	Blood			Derma			Organ-C			Organ-S			Ultrasound		
Method	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.3345	0.2467	0.2822	0.6291	0.4700	0.6244	0.2261	0.1772	0.1396	0.2864	0.1794	0.2475	0.2435	0.1581	0.1542
ties	0.2601	0.1872	0.1787	0.5995	0.4841	0.6484	0.2650	0.1684	0.2118	0.1981	0.1915	0.1874	0.3477	0.1989	0.1854
dare_linear	0.2832	0.2856	0.2392	0.6357	0.4841	0.6288	0.2018	0.2588	0.1650	0.2545	0.2245	0.2537	0.2162	0.1728	0.1608
dare_ties	0.3194	0.1935	0.1780	0.5375	0.4825	0.5558	0.1851	0.1515	0.1859	0.1790	0.1352	0.1730	0.2932	0.1635	0.1890
regmean	0.3460	0.2369	0.3407	0.5812	0.2843	0.5022	0.2405	0.1340	0.1388	0.2504	0.1623	0.1493	0.2276	0.1695	0.1318
fisher	0.3196	0.2237	0.2608	0.6707	0.6702	0.6690	0.2973	0.1741	0.1812	0.2010	0.2956	0.1963	0.2333	0.1860	0.2513
breadcrumbs	0.0854	0.1417	0.1460	0.0495	0.2913	0.1721	0.0766	0.0676	0.0685	0.0802	0.0948	0.0603	0.1225	0.1273	0.1021
model_stock	0.2420	0.1824	0.1975	0.6216	0.4311	0.5721	0.1225	0.1274	0.1185	0.2742	0.1578	0.2012	0.1791	0.1321	0.1578
from	0.2989	0.2306	0.2052	0.6713	0.4369	0.6580	0.2350	0.1560	0.1520	0.2207	0.2087	0.2476	0.2309	0.1506	0.2063
iso_c	0.3278	0.2748	0.3023	0.5049	0.4422	0.6416	0.2331	0.1629	0.1383	0.2736	0.2118	0.2453	0.2615	0.1698	0.1554
free_merge	0.3345	0.2467	0.2822	0.6291	0.4700	0.6246	0.2261	0.1771	0.1396	0.2863	0.1793	0.2475	0.2435	0.1581	0.1545
robustmerge	0.3401	0.2430	0.2256	0.2956	0.2085	0.3942	0.2458	0.1764	0.1653	0.2680	0.2227	0.2301	0.2189	0.2231	0.1707
LAMP-Merge	0.8002	0.7999	0.8007	0.6745	0.6761	0.6763	0.6074	0.6069	0.6076	0.5531	0.5524	0.5517	0.4762	0.4762	0.4762

Table 3: Client-average test accuracy on convnext. Each entry averages the three Dirichlet settings for a fixed client number.

	Blood			Derma			Organ-C			Organ-S			Ultrasound		
Method	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1565	0.1122	0.1384	0.2966	0.4830	0.6688	0.1037	0.1218	0.0944	0.1308	0.1035	0.1416	0.1506	0.1506	0.1252
ties	0.1715	0.1214	0.1524	0.2966	0.1797	0.2713	0.1217	0.0786	0.0716	0.1123	0.0761	0.0796	0.1518	0.1456	0.1291
dare_linear	0.1756	0.1122	0.1345	0.2705	0.4830	0.4830	0.1035	0.0699	0.1085	0.1282	0.1035	0.0797	0.1575	0.1575	0.1264
dare_ties	0.1096	0.1519	0.1497	0.0908	0.2798	0.2451	0.1305	0.1346	0.0770	0.1096	0.0946	0.0709	0.1456	0.1297	0.1500
regmean	0.1565	0.0817	0.1856	0.2633	0.0773	0.2720	0.1143	0.1017	0.0768	0.1014	0.1259	0.1038	0.1575	0.1539	0.1500
fisher	0.1565	0.0999	0.1189	0.4825	0.2638	0.2971	0.1003	0.0974	0.0915	0.0604	0.0517	0.0843	0.1695	0.1360	0.1456
breadcrumbs	0.1516	0.1366	0.1163	0.0524	0.2766	0.4630	0.1031	0.1444	0.1574	0.1516	0.1554	0.2077	0.1384	0.1363	0.1363
model_stock	0.1756	0.1144	0.1778	0.2966	0.4830	0.6688	0.1276	0.1218	0.1385	0.1308	0.1035	0.1935	0.1575	0.1291	0.1479
from	0.1565	0.1083	0.1671	0.2966	0.3406	0.4569	0.1232	0.1224	0.0650	0.1324	0.1105	0.0897	0.1575	0.1578	0.1087
iso_c	0.1947	0.1865	0.1480	0.2971	0.4830	0.6688	0.0991	0.0699	0.0945	0.0789	0.1477	0.0867	0.1518	0.1291	0.1605
free_merge	0.1575	0.1190	0.1384	0.2971	0.4830	0.6688	0.0906	0.1218	0.1385	0.1308	0.1035	0.1416	0.1506	0.1291	0.1695
robustmerge	0.1410	0.1405	0.0752	0.0908	0.4507	0.4830	0.1394	0.1218	0.1793	0.2354	0.0961	0.2077	0.1300	0.1135	0.1363
LAMP-Merge	0.8482	0.8483	0.8493	0.5992	0.5920	0.5852	0.6482	0.6480	0.6495	0.5977	0.5985	0.5990	0.4259	0.4259	0.4259

the same prevalence calibration to ordinary weight averaging. This comparison separates class-wise diagnostic reconstruction from prevalence calibration and tests whether the calibration term is effective without the diagnostic prototype geometry.

For the internal ablation, we use the representative long-tail stress case dermamnist_224/ResNet/$K=3/\beta=0.1$. For M1, we vary s, the scale of the reconstructed prototype head, to examine whether the classifier depends on a narrow logit magnitude. For M2, we vary λ, the bounded log-prior calibration strength. The activation threshold τ is fixed and only determines whether M2 is enabled. Table 12 shows that both components are stable over a dense local grid, with accuracy varying by less than one percentage point in the evaluated ranges.

Figure 1 visualizes the same grid as Table 15. The default s and λ lie in the high-accuracy plateau, while excessively strong M2 calibration slightly degrades accuracy. This supports treating prevalence calibration as a bounded correction rather than an unrestricted majority-class prior.

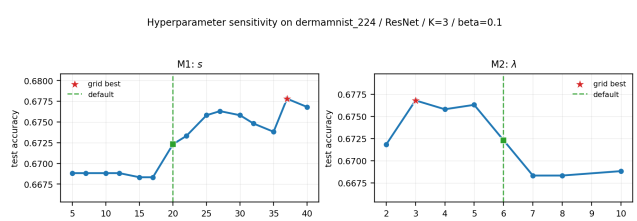

Figure 1: Hyperparameter sensitivity of M1 and M2 on dermamnist_224/ResNet/$K=3/\beta=0.1$. Red stars denote grid-best values and green dashed lines denote the default values used by the formal method ($s=20$, $\lambda=6$).

Table 4: Client-average test accuracy on vit_t. Each entry averages the three Dirichlet settings for a fixed client number.

	Blood			Derma			Organ-C			Organ-S			Ultrasound		
Method	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1343	0.1082	0.1160	0.4638	0.4369	0.3696	0.1657	0.1043	0.0952	0.0844	0.0852	0.1710	0.1450	0.1327	0.1884
ties	0.0796	0.1421	0.1285	0.4780	0.3498	0.4615	0.1341	0.1223	0.0953	0.1192	0.1002	0.1597	0.2001	0.1306	0.1018
dare_linear	0.2397	0.1129	0.1387	0.2850	0.1002	0.1204	0.1534	0.0946	0.0704	0.0748	0.0549	0.1611	0.1366	0.1051	0.1435
dare_ties	0.0735	0.1189	0.1146	0.2735	0.4502	0.3528	0.1252	0.0903	0.1033	0.0541	0.0874	0.0591	0.1447	0.1120	0.1042
regmean	0.1748	0.1417	0.1290	0.5114	0.3667	0.6658	0.1250	0.0964	0.0983	0.1263	0.0717	0.1068	0.1491	0.1384	0.1144
fisher	0.1166	0.1512	0.1944	0.4780	0.2042	0.4820	0.0799	0.1072	0.1424	0.0856	0.1218	0.1222	0.1611	0.1315	0.1420
breadcrumbs	0.1363	0.1039	0.1300	0.0489	0.2572	0.0607	0.0773	0.1000	0.1644	0.1237	0.0483	0.1219	0.1360	0.1141	0.1524
model_stock	0.1940	0.1120	0.1134	0.3805	0.3079	0.2735	0.1488	0.1227	0.1215	0.0893	0.0859	0.1867	0.1599	0.1809	0.1641
from	0.1802	0.0781	0.1405	0.4695	0.5121	0.6688	0.2148	0.1301	0.0972	0.0772	0.0798	0.1485	0.1836	0.1743	0.1776
iso_c	0.1234	0.1483	0.1018	0.5709	0.4529	0.2306	0.1084	0.0567	0.0897	0.0611	0.0531	0.1081	0.1417	0.1527	0.1869
free_merge	0.1989	0.1457	0.1392	0.4630	0.4770	0.3855	0.1425	0.0980	0.0759	0.0727	0.0914	0.1643	0.1405	0.1593	0.1647
robustmerge	0.1482	0.1070	0.1425	0.0347	0.2768	0.1724	0.0954	0.0617	0.1438	0.1260	0.0707	0.1217	0.1698	0.1854	0.1563
LAMP-Merge	0.8523	0.8522	0.8516	0.5970	0.5938	0.5860	0.5805	0.5790	0.5790	0.5406	0.5372	0.5388	0.4645	0.4645	0.4645

Table 5: Client-average test accuracy on swin_tiny. Each entry averages the three Dirichlet settings for a fixed client number.

	Blood			Derma			Organ-C			Organ-S			Ultrasound		
Method	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1565	0.1186	0.1524	0.4569	0.4830	0.6688	0.1068	0.1294	0.0787	0.0682	0.0604	0.1381	0.1527	0.1294	0.1465
ties	0.0972	0.1190	0.1384	0.0841	0.0780	0.6688	0.1160	0.0702	0.0934	0.0734	0.0607	0.1107	0.1695	0.1303	0.1261
dare_linear	0.1565	0.1161	0.1825	0.4569	0.2771	0.4630	0.1155	0.1272	0.0776	0.0640	0.0720	0.1381	0.1743	0.1536	0.1497
dare_ties	0.1384	0.1162	0.1384	0.2961	0.2638	0.6688	0.1276	0.0774	0.1116	0.1194	0.0434	0.1230	0.1791	0.1351	0.1300
regmean	0.1374	0.1715	0.1578	0.2705	0.4630	0.4569	0.1411	0.0773	0.0811	0.1215	0.0821	0.0926	0.1465	0.1105	0.1779
fisher	0.1633	0.1756	0.1565	0.4582	0.0652	0.4718	0.0814	0.0899	0.0978	0.0651	0.0650	0.1092	0.1746	0.1165	0.1755
breadcrumbs	0.1500	0.1365	0.1537	0.0647	0.2771	0.0715	0.1445	0.1783	0.1660	0.2354	0.2077	0.2111	0.1450	0.1312	0.1653
model_stock	0.1374	0.1440	0.1715	0.4569	0.2771	0.4825	0.1110	0.0771	0.1404	0.0671	0.1106	0.1714	0.1539	0.1273	0.1869
from	0.1565	0.1495	0.1671	0.4569	0.2971	0.6688	0.1174	0.1220	0.1051	0.1133	0.1016	0.1380	0.1539	0.1252	0.1476
iso_c	0.1430	0.1722	0.1700	0.3099	0.2840	0.5174	0.1302	0.1327	0.1469	0.1545	0.1727	0.1880	0.1755	0.1731	0.1791
free_merge	0.1154	0.0861	0.1840	0.4569	0.4830	0.6688	0.0959	0.1377	0.0890	0.0682	0.0497	0.1062	0.1518	0.1575	0.1599
robustmerge	0.1374	0.1409	0.1670	0.2377	0.2771	0.0504	0.1601	0.1442	0.1960	0.1673	0.1151	0.2077	0.1761	0.1315	0.1629
LAMP-Merge	0.7737	0.7753	0.7748	0.6374	0.6251	0.6244	0.6651	0.6632	0.6630	0.6078	0.6076	0.6068	0.4816	0.4816	0.4816

V CONCLUSION

We studied one-shot asynchronous medical model merging, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical analysis shows that generic post-hoc merging methods suffer from aggregation-induced predictive collapse under incomplete local diagnostic coverage. We further show that this collapse can be masked by raw accuracy on long-tailed medical datasets, motivating evaluation with macro recall, macro F1, and prediction-distribution diagnostics.

LAMP-Merge addresses these two properties with a concise two-module design. M1 reconstructs a class-wise diagnostic prototype head from client-side class prototypes in a shared reference feature space, preserving local evidence for each diagnosis class. M2 applies bounded prevalence calibration when uploaded class counts indicate a strongly long-tailed medical prior. Across five medical datasets and five backbone families, LAMP-Merge wins or ties the strongest non-LAMP baseline in 68 out of 75 client-average cells, while also improving balanced accuracy, macro F1, collapse ratio, and effective predicted classes in the diagnostic analysis. These results indicate that medical class-level statistics provide a practical and privacy-preserving signal for asynchronous clinical model fusion.

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Table 6: Client-average test accuracy on openai/clip-vit-base-patch32. Each entry averages the three Dirichlet settings for a fixed client number.

	Blood			Derma			Organ-C			Organ-S			Ultrasound		
Method	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$	$K=3$	$K=5$	$K=7$
avg	0.1857	0.1199	0.1863	0.6688	0.6688	0.5682	0.2360	0.1348	0.1226	0.2978	0.0926	0.1447	0.1602	0.1282	0.1087
ties	0.2472	0.2286	0.1911	0.6677	0.6760	0.4906	0.2868	0.1682	0.1728	0.2797	0.2139	0.1847	0.1668	0.1204	0.1264
dare_linear	0.1715	0.1202	0.1773	0.6688	0.6688	0.5644	0.2203	0.1342	0.1286	0.3028	0.0863	0.1365	0.1518	0.1312	0.1093
dare_ties	0.2588	0.2306	0.1616	0.6717	0.6756	0.4934	0.2498	0.1618	0.1586	0.2796	0.2063	0.1791	0.1752	0.1204	0.1441
regmean	0.2740	0.2043	0.1907	0.6710	0.5520	0.5104	0.2764	0.1924	0.1394	0.3468	0.2295	0.1853	0.1506	0.1423	0.1087
fisher	0.1819	0.1384	0.1604	0.6715	0.4830	0.4828	0.2910	0.1753	0.1463	0.2397	0.0967	0.1403	0.1387	0.1087	0.1288
breadcrumbs	0.1459	0.1202	0.1170	0.6688	0.6657	0.4894	0.1816	0.1236	0.0766	0.2034	0.0954	0.0768	0.1207	0.1330	0.1731
model_stock	0.1689	0.1163	0.1947	0.6688	0.6687	0.5372	0.2121	0.1844	0.0661	0.2550	0.0861	0.1231	0.1539	0.1363	0.1695
from	0.2727	0.1618	0.1942	0.6715	0.6780	0.6399	0.2491	0.1832	0.1943	0.2435	0.2243	0.1886	0.1758	0.1360	0.1264
iso_c	0.1222	0.1303	0.1137	0.6369	0.2118	0.4698	0.1165	0.1050	0.1181	0.0762	0.1004	0.0831	0.1255	0.1479	0.1692
free_merge	0.2145	0.1141	0.2349	0.6688	0.6683	0.5623	0.1501	0.1419	0.0816	0.2489	0.1136	0.0707	0.1488	0.1411	0.1186
robustmerge	0.1680	0.1575	0.1947	0.6688	0.6680	0.4921	0.2487	0.1256	0.1200	0.2701	0.0677	0.1556	0.1575	0.1369	0.1539
LAMP-Merge	0.3908	0.2679	0.2483	0.5646	0.6846	0.6688	0.4460	0.2242	0.1468	0.2670	0.2296	0.2558	0.1947	0.1608	0.1602

Table 7: Prediction-collapse diagnostics averaged over five medical datasets and four backbones per dataset.

Metric	LAMP-Merge	Best reference	Difference
Accuracy ↑	0.5885	0.3156	+0.2729
BA / macro recall ↑	0.5747	0.1810	+0.3937
Macro F1 ↑	0.5352	0.1123	+0.4229
Collapse ratio ↓	0.2294	0.7477	-0.5184
Effective classes ↑	8.0390	2.1360	+5.9030
Pred-true TV ↓	0.1510	0.6168	-0.4658

Table 8: Client-average module ablation over 75 cells.

Setting	Mean Acc	Best/tied	$\geq$ avg	Margin
LAMP-Merge	0.5618	51/75	71/75	+0.3345
M1 only	0.5362	46/75	66/75	+0.3089
avg+M2	0.2198	4/75	43/75	-0.0075
avg	0.2273	2/75	75/75	0.0000

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Table 9: Dataset-level client-average accuracy in the module ablation. Each dataset contains 15 cells from five model families and three client counts; each cell is averaged over three Dirichlet skew levels.

Dataset	Cells	LAMP-Merge	M1 only	avg+M2	avg	LAMP best/tied
Blood	15	0.7156	0.7155	0.1726	0.1699	12/15
Ultrasound	15	0.4040	0.4040	0.1551	0.1516	15/15
Derma	15	0.6257	0.4975	0.4913	0.5304	12/15
Organ-C	15	0.5543	0.5543	0.1331	0.1358	6/15
Organ-S	15	0.5096	0.5096	0.1469	0.1488	6/15

Table 10: Inter-module ablation settings.

Setting	M1	M2	Controlled hypothesis
LAMP-Merge	yes	yes	complete design
M1 only	yes	no	prototype contribution
avg+M2	no	yes	prior calibration alone
avg	no	no	parameter averaging control

Table 11: Client-average inter-module ablation over the full benchmark.

Setting	Cells	Mean Acc	Best/tied	$\geq$ avg
LAMP-Merge	75	0.5618	51/75	71/75
M1 only	75	0.5362	46/75	66/75
avg+M2	75	0.2198	4/75	43/75
avg	75	0.2273	2/75	75/75

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Table 12: Internal ablation on the long-tail stress case.

Module	Factor	Value	Acc
M2	λ	2	0.6718
M2	λ	3	0.6768
M2	λ	4	0.6758
M2	λ	5	0.6763
M2	λ	6	0.6723
M2	λ	7	0.6683
M2	λ	8	0.6683
M2	λ	10	0.6688
M1	s	5	0.6688
M1	s	7	0.6688
M1	s	10	0.6688
M1	s	12	0.6688
M1	s	15	0.6683
M1	s	17	0.6683
M1	s	20	0.6723
M1	s	22	0.6733
M1	s	25	0.6758
M1	s	27	0.6763
M1	s	30	0.6758
M1	s	32	0.6748
M1	s	35	0.6738
M1	s	37	0.6778
M1	s	40	0.6768

Table 13: Metric diagnostics averaged over BloodMNIST, Ultrasound, DermaMNIST, Organ-C, and Organ-S, with four backbones per dataset. Lower is better for ρ and TV; higher is better for the other metrics.

Method	Acc	BA	F1	ρ	C_{eff}	TV
client mean	0.1804	0.1515	0.0638	0.8207	1.7370	0.7644
client best	0.3156	0.1810	0.1123	0.7619	2.0201	0.6168
avg	0.2304	0.1543	0.0751	0.8028	1.7666	0.7082
ties	0.2089	0.1682	0.0858	0.7477	2.1360	0.6907
dare_linear	0.2145	0.1481	0.0745	0.7577	2.0556	0.6849
fisher	0.2585	0.1604	0.0886	0.8228	1.9324	0.6671
from	0.2494	0.1652	0.0848	0.7705	1.8889	0.6949
iso_c	0.1976	0.1648	0.0845	0.7538	2.1161	0.7172
robustmerge	0.1486	0.1425	0.0502	0.8225	1.6559	0.8035
LAMP-Merge	0.5885	0.5747	0.5352	0.2294	8.0390	0.1510

Table 14: Dataset-level metric diagnostics. Each row compares LAMP-Merge with the strongest non-LAMP/client reference under the same metric. Lower is better for ρ and TV.

Dataset	LAMP Acc	Ref Acc	LAMP BA	Ref BA	LAMP F1	Ref F1	LAMP ρ	Ref ρ	LAMP TV	Ref TV
Blood	0.8190	0.2286	0.8071	0.2129	0.8022	0.1133	0.1943	0.7768	0.0394	0.6759
Ultrasound	0.4625	0.2682	0.4536	0.2335	0.4378	0.1514	0.2035	0.6047	0.1575	0.6094
Derma	0.4627	0.6726	0.4492	0.1890	0.2934	0.1562	0.3697	0.7978	0.3309	0.3185
Organ-C	0.6249	0.2297	0.6175	0.1708	0.6030	0.1067	0.1772	0.5474	0.1280	0.6055
Organ-S	0.5734	0.2586	0.5460	0.1603	0.5397	0.0881	0.2021	0.6274	0.0993	0.6226

Table 15: Hyperparameter sensitivity summary on dermamnist_224/ResNet/$K = 3/\beta = 0.1$.

Module	Parameter	Best	Best Acc	Worst	Range
M1	s	37	0.6778	15	0.0095
M2	λ	3	0.6768	7	0.0085